MA 104: Ordinary Differential Equations

Lecture 1

January 7, 2026

Tutorials and Lectures

Lecture Day and Time:

- Wednesday & Friday (10:00 AM – 11:20 AM)

Tutorial Day and Time:

- Wednesday (11:30 AM – 12:50 AM)

Office Hour:

- Monday (08:30 AM – 09:30 AM)
- Office No.: AB 06/347 (by appointment)

Google Classroom Code: vmileaxf

Suggestions and comments are welcome throughout the course!

Grading Policy

Relative grading policy will be followed.

Evaluation Component	Weightage (%)
Quiz 1 (Jan 27, tentative)	20
Quiz 2 (Feb 16, tentative)	20
Final Examination	40
Assignment	10
Attendance	10
Total	100

Attendance (%)	Marks
85% and above	10
70% to less than 85%	8
55% to less than 70%	6
40% to less than 55%	4
Below 40%	0

References

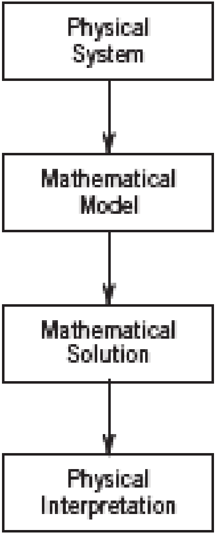
Textbook:

- E. Kreyszig, *Advanced Engineering Mathematics* (10th Edition), John Wiley, 2011.

Reference books:

- W. E. Boyce and R. DiPrima, *Elementary Differential Equations* (8th Edition), John Wiley, 2005.
- G.F. Simmons, *Differential Equations with Applications and Historical Notes*, Tata McGraw-Hill Publishing Company Limited, 1974.

Basic Concepts: Modeling



Ordinary Differential Equations

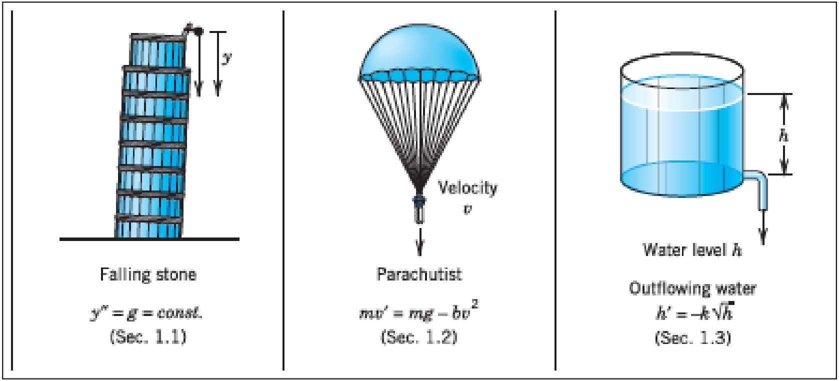
An ordinary differential equation (ODE) is an equation that contains one or more derivatives of an unknown function, usually denoted by $y(x)$ (or $y(t)$ when time t is the independent variable).

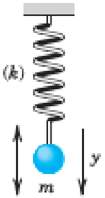
Examples.

- $y'' = x^3 + y$,
- $y'' = \sin(x + y)$,
- $y^{(4)} = (x - 4)^2 + x^5$.

An ODE is said to be of order n if the n^{th} derivative of the unknown function y is the highest derivative of y in the equation.

Some Applications of Differential Equations



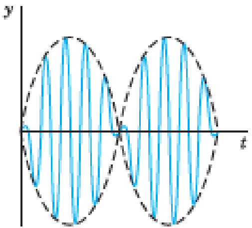


Displacement y

Vibrating mass on a spring

$my'' + ky = 0$

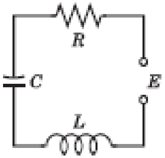
(Secs. 2.4, 2.8)



Beats of a vibrating system

$y'' + \omega_0^2 y = \cos\ \omega t, \quad \omega_0 \approx \omega$

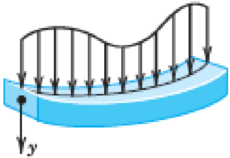
(Sec. 2.8)



Current I in an RLC circuit

$LI'' + RI' + \frac{1}{C} I = E'$

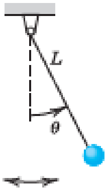
(Sec. 2.9)



Deformation of a beam

$EIy^{iv} = f(x)$

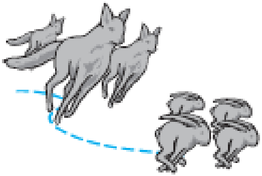
(Sec. 3.3)



Pendulum

$L\theta'' + g \sin\theta = 0$

(Sec. 4.5)



Lotka–Volterra predator–prey model

$y_1' = ay_1 - by_1y_2$

$y_2' = ky_1y_2 - ly_2$

(Sec. 4.5)

First Order ODEs

Equations of this type involve only the first derivative and may include y along with given functions of x. Hence they can be expressed as:

Implicit form:

$$\boxed{F(x, y, y') = 0.}$$

Explicit form:

$$\boxed{y' = f(x, y).}$$

A function $y = h(x)$ is called a solution of $F(x, y, y') = 0$ on $a < x < b$ if:

- h is differentiable on (a, b),
- substitution of $y = h(x)$, $y' = h'(x)$ gives an identity.

The graph of h is called a solution curve or integral curve.

Do solutions always exist?

- Do all ODEs have solutions?
- If yes, how many?

Example

The ODE

$$|y'| + |y| + 1 = 0$$

admits no real (or complex) valued function y as a solution, because the sum of positive quantities cannot be a negative number.

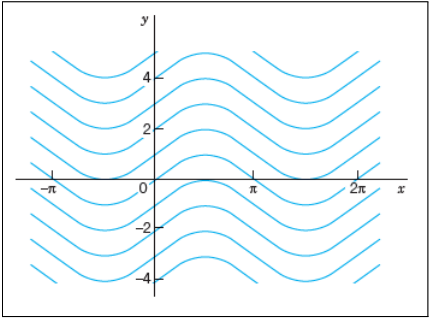
Examples

⋆ Verify: $\phi(x) = \frac{c}{x}$ (c arbitrary) is a solution of $xy' = -y$, $x \neq 0$.

⋆ For the ODE $y' = \cos x$, using calculus we obtain

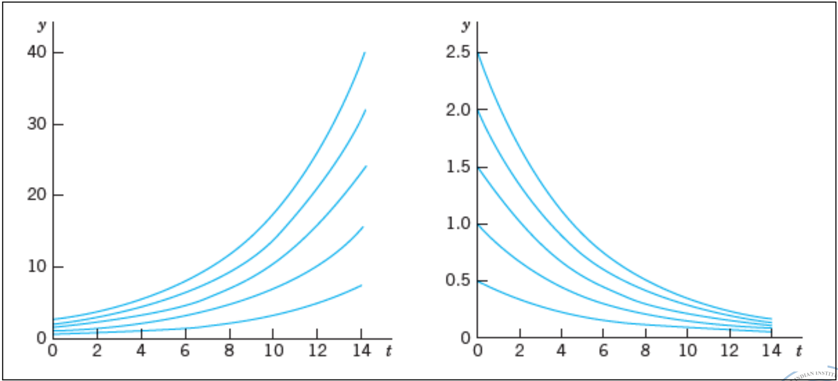
$$y = \int \cos x \, dx = \sin x + c,$$

the family of solutions. For $c = -3, -2, -2, 0, 1, 2, 3,$



Growth and Decay Curves

Consider $y' = ky$. This has solutions $y = ce^{kt}$, modeling exponential growth (Malthus' law) if $k > 0$ and exponential decay if $k < 0$.



Solutions Outside Family of Solutions

In the earlier examples, the ODEs had unique families of solutions. However, this may not always be the case.

Example. Consider the first-order ODE

$$(y')^2 - 4y = 0.$$

The one-parameter family of solutions

$$y_c(x) := (x + c)^2$$

satisfies the ODE. In addition, $y \equiv 0$ is also a solution, which is not included in the above family of solutions.

• A particular solution is obtained from the general solution using an initial condition $y(x_0) = y_0$, which determines the constant C.

Geometrically, the solution curve should pass through (x_0, y_0).

• An ODE $y' = f(x, y)$ together with an initial condition $y(x_0) = y_0$ is called an initial value problem (IVP).

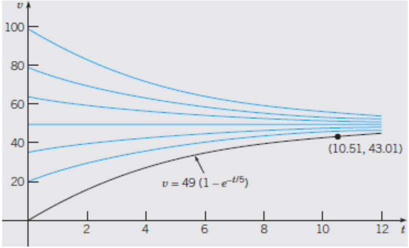


Figure: $v' = 9.8 - v/5$ with $v(0) = 0$

Examples

A first-order IVP may exhibit the following behaviors:

1. No solution: $|y'| + |y| = 0, \quad y(0) = 3.$
2. A unique solution: $y' = x, \quad y(0) = 1.$ (Find the solution!)
3. Infinitely many solutions: $xy' = y - 1, \quad y(0) = 1.$
   The family of solutions is $y = 1 + cx$.

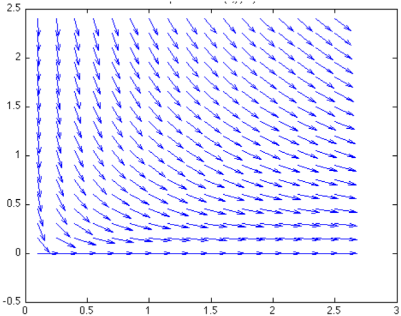
These examples motivate the study of conditions under which solutions to an IVP *exist* and are *unique*.

Geometrical meaning: dy/dx=-y/x

Find the curve through the point $(1, 1)$ in the xy-plane having, at each of its points, the slope $-y/x$.

$\star$ Recall: $\phi(x) = c/x$ (c arbitrary) is a solution of the relevant ODE is $y' = -y/x$ with $x \neq 0$.

The initial condition given is $y(1) = 1$, which implies $c = 1$. Hence the particular solution for the above problem is $y = 1/x$.



MA 104

Lecture 1

January 7, 2026

1st order ODE (explicit form)

$$y' = f(x,y)$$

Suppose f is a fn only in x. So we can write

$$\boxed{\frac{dy}{dx} = f(x)}$$

$$\Rightarrow \quad y = \int f(x)\,dx + c := h_c(x).$$

Rmk. The indefinite integral can be solved by the methods of calculus. In certain cases it may be difficult/impossible (Eg. $\int e^{-x^2}dx$, $\int \frac{\sin x}{x}\,dx$).

$$\frac{dy}{dx} = f(x,y) := -\frac{M(x,y)}{N(x,y)} \quad \left\{ \boxed{\begin{array}{l} M(x,y) = f(x,y) \\ N(x,y) = -1 \end{array}} \right.$$

$$\Rightarrow \quad \boxed{M(x,y)\,dx + N(x,y)\,dy = 0} \quad -(1)$$

(Eg. $f(x,y) = x/y \sim M(x,y) = -x$, $N(x,y) = y$)

$M(x,y) = M(x)$, $N(x,y) = N(y)$

Rewriting (1) $\Rightarrow$ $\boxed{M(x)\,dx + N(y)\,dy = 0}$

$$\Rightarrow \quad \int M(x)\,dx = -\int N(y)\,dy$$

seperation of variables!

Eg.(1) $\underline{y' = 1 + y^2}$ $\rightsquigarrow$ $M(x)\,dx + N(y)\,dy = 0$

$\boxed{M(x) = 1}$, $N(y) = -\frac{1}{1+y^2}$.

$dy = (1+y^2)\,dx$

$\Rightarrow \quad dx - \frac{1}{1+y^2}\,dy = 0$

$\Rightarrow \quad \int \frac{1}{1+y^2}\,dy = \int dx + c$

$\Rightarrow \quad \tan^{-1}(y) = x + c$

$\Rightarrow \quad \boxed{y = \tan(x + c)}$.

(2) $y' = -2xy$, $\boxed{y(0) = 1.8}$

$\int \frac{dy}{y} = \int -2x\,dx$

$\Rightarrow \ln y = -x^2 + \ln c$

$\Rightarrow \boxed{y = ce^{-x^2}}$ $\rightsquigarrow$ $\boxed{y = (1.8)e^{-x^2}}$

Given; $1.8 = y(0) = ce^0 = c \Rightarrow c = 1.8$.

A fun^n $f(x,y)$ is called homogeneous of degree n

if $\boxed{f(tx, ty) = t^n f(x,y)}$

Eg. $f(x,y) = \underline{x^a y^b}$, $\underline{f(tx,ty)} = t^{a+b} x^a y^b = t^{a+b} f(x,y)$

Eg. x^2+y^2+xy, $\cos(y/x)$, $\sqrt{x^2+y^2}$.
deg 2 deg 0 deg 1

$M(x,y)$ & $N(x,y)$ $\rightsquigarrow$ both are homog. of same degree

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{M(x,y)}{N(x,y)} := \underline{f(x,y)}$$

homog. of deg 0.

$$f(x,y) = f\left(x \cdot 1,\ x \cdot y/x\right) = x^0 f(1, y/x)$$

$$= f(1, y/x) := \tilde{f}(y/x)$$

$\rightarrow$ Consider $\boxed{y' = g(y/x)}$ $\rightsquigarrow$ ODE of this form

$u = y/x \rightsquigarrow \underline{y = ux}$

$y' = u'x + u \rightsquigarrow$ product formula.

$\rightarrow$ Rewriting: $u'x + u = g(u)$

$$\Rightarrow u'x = g(u) - u$$

$$\Rightarrow \int \frac{du}{g(u) - u} = \int \frac{dx}{x} + c$$

$$\Rightarrow u = \varphi_c(x)$$

$$y = ux = x\,\varphi_c(x)$$

Eg. 1 $\quad 2xy\,y' = y^2 - x^2$

$$\Rightarrow (x^2 - y^2)\,dx + 2xy\,dy = 0$$

$M(x,y) = x^2 - y^2$, $N(x,y) = 2xy$ $\rightsquigarrow$ homog. of same deg 2.

$$\Rightarrow y' = \frac{y^2 - x^2}{2xy} = \frac{y}{2x} - \frac{x}{2y}$$

$$\left.\begin{array}{l} u = \frac{y}{x} \\ y = xu \\ y' = xu' + u \end{array}\right\} \Rightarrow xu' + u = \frac{u}{2} - \frac{1}{2u}$$

$$\Rightarrow xu' = -\frac{u}{2} - \frac{1}{2u} = -\frac{u^2+1}{2u}$$

$$\Rightarrow \int \frac{2u}{u^2+1}\,du = \int -\frac{dx}{x}$$

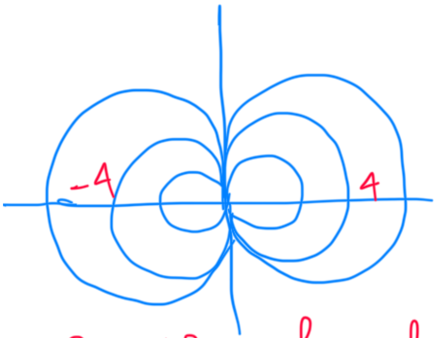
$$\Rightarrow \ln(u^2+1) = -\ln|x| + \ln c$$

$$\Rightarrow u^2 + 1 = c/x$$

$$\Rightarrow (y/x)^2 + 1 = c/x$$

$$\Rightarrow x^2 + y^2 = cx$$

$$\Rightarrow (x - c/2)^2 + y^2 = c^2/4$$



Family of solu^ns.

Exact Equ^ns

Eg: $\quad 2x + y^2 + 2xy\,y' = 0$

$$\Rightarrow \underbrace{(2x+y^2)}_{\boxed{M(x,y)}}\,dx + \underbrace{2xy}_{N(x,y)}\,dy = 0 \quad - (1)$$

NOT homog. $\Leftarrow$

Take $u(x,y) = x^2 + xy^2$

$$\frac{\partial u}{\partial x} = 2x + y^2 = M(x,y), \quad \frac{\partial u}{\partial y} = 2xy = N(x,y)$$

So (1) $\Rightarrow \dfrac{\partial u}{\partial x}\,dx + \dfrac{\partial u}{\partial y}\,dy = 0$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx} = 0 \quad - (2)$$

Notice $u(x, y(x))$ is a fu^n in x & $y(x)$.

So by the chain rule,

$$\boxed{\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}}$$

Hence ② $\Rightarrow$ $\frac{du}{dx} = 0$ $\Rightarrow$ $\boxed{u = c}$.

Thus the solu^n s of ① are given implicitely by $u(x, y(x)) = c$, for an arbitrary const. c.